

Can isotopes help us better understand clouds?

Clouds play a crucial role in the Earth's climate and water cycle. They regulate the distribution of rainfall, influence weather patterns, and affect how much solar energy is absorbed or reflected by the atmosphere. Despite their importance, many processes occurring inside clouds remain difficult to observe directly and are still poorly represented in weather and climate models.

One possible way to study these processes is through stable water isotopes. Water molecules are not all identical: some contain slightly heavier forms of hydrogen or oxygen. These differences cause isotopes to behave differently during evaporation, condensation, freezing, and melting. As a result, the isotopic composition of atmospheric water carries information about the processes that have shaped a cloud during its lifetime.

The aim of this project is to investigate whether isotopic measurements can be used to better understand how clouds evolve. To achieve this, a new computer model will be developed that simulates the growth and evolution of individual cloud droplets and ice particles while simultaneously tracking their isotopic composition. The model will be compared with measurements collected during aircraft campaigns and laboratory experiments.

A key question addressed by the project is whether isotopic observations contain enough information to explain the isotopic state of a cloud in a physically consistent way. Answering this question will help determine which cloud processes leave measurable isotopic signatures and whether isotopes can provide information about cloud properties that are otherwise difficult to observe.

The project combines modern cloud modelling with atmospheric isotope research, two fields that have rarely been brought together at this level of detail. The expected outcome is a new modelling framework for studying isotopes in clouds and a better understanding of the information contained in isotopic measurements. In the longer term, the results may contribute to improving the representation of clouds in atmospheric models and advancing our understanding of the Earth's water cycle.